

**Paper Practice 9.5****Geometric Sequences**

**WORKED KEY – Show all of your work in the space provided. Use  $a_n = a_1 \cdot r^{(n-1)}$  where a formula is needed. Round money to the nearest whole unit.**

- 1.** Determine whether each sequence is geometric. If it is, state the common ratio. (Is It Geometric?)
  - a.**  $-432, 144, -48, 16, \dots$       ANSWER: Geometric,  $r = -\frac{1}{3}$
  - b.**  $4, 9, 25, 36, \dots$       ANSWER: Not geometric –  $9/4 \neq 25/9$ .
  - c.**  $16, 12, 8, 4, \dots$       ANSWER: Not geometric – the terms drop by a constant 4, so it is arithmetic.
- 2.** Find the next three terms in each geometric sequence. (Next Terms)
  - a.**  $64, 16, 4, 1, \dots$       ANSWER:  $r = \frac{1}{4}$ :  $\frac{1}{4}, \frac{1}{16}, \frac{1}{64}$
  - b.**  $8, 12, 18, 27, \dots$       ANSWER:  $r = \frac{3}{2}$ :  $40.5, 60.75, 91.125$
  - c.**  $3, -6, 12, -24, \dots$       ANSWER:  $r = -2$ :  $48, -96, 192$
- 3.** Find the next three terms in each geometric sequence. (Next Terms)
  - a.**  $96, 48, 24, 12, \dots$       ANSWER:  $r = \frac{1}{2}$ :  $6, 3, 1.5$
  - b.**  $2, 10, 50, 250, \dots$       ANSWER:  $r = 5$ :  $1250, 6250, 31250$
- 4.** Write an explicit formula  $a_n = a_1 \cdot r^{(n-1)}$  for each sequence. (Explicit Formula)
  - a.**  $5, 15, 45, 135, \dots$       ANSWER:  $a_n = 5(3)^{(n-1)}$
  - b.**  $800, 400, 200, 100, \dots$       ANSWER:  $a_n = 800(\frac{1}{2})^{(n-1)}$
- 5.** A sequence has  $a_1 = 6$  and  $r = -2$ . Write the first five terms. (Build a Sequence)      ANSWER:  $6, -12, 24, -48, 96$

- 6.** Use an explicit formula to find the indicated term. (Find a Term)

**a.** 512, 256, 128, 64, ... find  $a_{11}$

ANSWER:  $r = \frac{1}{2}$ ;  $a_{11} = 512(\frac{1}{2})^{10} = 512/1024 = \frac{1}{2}$

**b.** -8, 24, -72, 216, ... find  $a_9$

ANSWER:  $r = -3$ ;  $a_9 = -8(-3)^8 = -8(6561) = -52,488$

- 7.** Use an explicit formula to find the indicated term. (Find a Term)

**a.** 5, 3, 1.8, 1.08, ... find  $a_8$       ANSWER:  $r = 0.6$ ;  $a_8 = 5(0.6)^7 = 0.139968$

**b.** 7, 21, 63, ... find  $a_6$       ANSWER:  $r = 3$ ;  $a_6 = 7(3)^5 = 7(243) = 1701$

- 8.** North Dakota's population was 685,242 in 2011 and has been increasing about 2.5% each year. Estimate the population in 2030.

(Apply)      ANSWER: 19 years:

$685242(1.025)^{19} \approx 1,095,462$  people

- 9.** Global vinyl record sales were \$267 million in 2014 and have been increasing about 26% each year. Estimate sales for 2025. (Apply)  
ANSWER: 11 years:  $267(1.26)^{11} \approx \$3,393$  million (about \$3.4 billion)

- 10.** Error Analysis – Gus says 2, 4, 6, 8, ... is geometric because the terms keep increasing. Identify his mistake. (Error Analysis)  
ANSWER: Increasing is not enough – a geometric sequence needs a constant RATIO. Here  $4/2 = 2$  but  $6/4 = 1.5$ . It is arithmetic, with a common difference of 2.

- 11.** Explain the difference between an arithmetic and a geometric sequence, and give an example of each. (Explain)      ANSWER: Arithmetic adds a common difference; geometric multiplies by a common ratio. Answers vary.